# Automated Quality Assessment and Adaptive Mentorship for Undergraduate Research Proposals

**INDIVIDUAL THESIS**

RASHMIKA R.M.D

IT22101624

Individual component: Automated Quality assessment and adaptive mentorship

Group ID: R26-IT-095

Report submitted in partial fulfilment of the requirements for the

Information Technology

Faculty of Computing

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

2026 October

# DECLARATION AND SUPERVISOR CERTIFICATION

Candidate statement. I declare that this thesis is my own work and that all sources used have been acknowledged. Except where acknowledged, no part has been previously published or written by another person. I grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this thesis, including through print and electronic transmission. I reserve the right to use its contents in future work. Generative AI assisted the organization, writing and formatting of this document, as disclosed in Appendix B. I remain accountable for the accuracy, originality and attribution of its content.

Student name: Rashmika R M D

Student ID: IT22101624

Signature: _____  _____ Date: 10/10/2026_____

**Supervisor certification**. The above candidate carried out this research for the bachelor's degree thesis under my supervision.

**Supervisor certification**

| Role and name | Signature | Date |
|---|---|---|
| Supervisor: [Name] | | |
| Co-supervisor: [If applicable] | | |

# ABSTRACT

Undergraduate research proposals require feedback on structure, methodological clarity and academic justification. A numerical assessment alone provides limited guidance for revision, while fully automated evaluation risks presenting uncertain predictions as authoritative judgements. This research develops a human-in-the-loop component for automated quality assessment and adaptive mentorship, combining proposal validation, weakness classification, historical feedback retrieval and learning resource recommendations within a supervisor-controlled workflow.

The solution uses a React interface, a FastAPI backend and PostgreSQL persistence to associate analysis and review records with proposal versions. The deployed weakness classifier combines all-MiniLM-L6-v2 sentence embeddings with balanced Logistic Regression and uses 2,121 deduplicated annotation spans. Historical feedback retrieval uses 2,151 embedded records. A separate TF-IDF and Ridge Regression pipeline estimates an advisory score from 165 grading records on a 0–42 scale. The selected

classification targets are accuracy of 85%, macro-F1 of 0.800 and weightedF1 of 0.850. The selected grading targets are MAE of 1.50 points, RMSE of 2.00 points and $R^2$ of 0.50. These values are performance targets requiring experimental verification; the grader remains advisory.

The reviewed API logic requires a saved supervisor review before the relevant final feedback artefacts can be generated. However, the recorded test attempt stopped during collection because of a missing dependency, and no user study establishes learning improvement or reduced automation bias. The contribution is an integrated, versionaware assessment and mentorship prototype with an explicit human review boundary. Further runtime verification and stakeholder evaluation are required before educational deployment.

Keywords: proposal evaluation, adaptive mentoring, sentence embeddings, humaninthe-loop, formative feedback

## ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my supervisor, **Prof. Samantha Rajapaksha**, and co-supervisor, **Ms. Kaushika Kahatapitiya**, for their valuable guidance, constructive feedback, and continuous support throughout this research project. Their advice was important in refining the research problem, methodology, evaluation process, and development of the Intelligent Team Formation and Topic Feasibility Analysis component.

I also extend my appreciation to the academic staff of the Sri Lanka Institute of Information Technology for providing the knowledge, resources, and research environment required to complete this work.

Thirdly, I would like to express my gratitude to my teammates for their cooperation and discussions during this process of research.

I would also like to thank my research group members for their cooperation in developing and integrating the complete **Automated Quality Assessment and Adaptive metorship**. Their collaboration was important in ensuring that the individual components could operate together as an integrated solution.

# Table of Contents


**1 INTRODUCTION ..... 9**
1.1 Overall project context ..... 9
1.2 Individual component context ..... 9
1.3 Problem statement ..... 9
1.4 Research gap ..... 10
1.5 Aim and objectives ..... 10
1.6 Research questions or hypotheses ..... 11
1.7 Scope limitations and assumptions ..... 11
1.8 Individual contribution ..... 11
1.9 Thesis structure ..... 12
**2 LITERATURE REVIEW AND RELATED WORK ..... 12**
2.1 Review method and scope ..... 12
2.2 Domain and theoretical foundations ..... 12
2.3 Related research ..... 13
2.4 Existing systems products and techniques ..... 14
2.5 Gap synthesis and conceptual framework ..... 14
**3 RESEARCH METHODOLOGY ..... 14**
3.1 Research design ..... 14
3.2 Data participants or study materials ..... 15
3.3 Experimental or study procedure ..... 16
3.4 Measures and evaluation metrics ..... 16
3.5 Data analysis ..... 17
3.6 Ethics privacy security and responsible AI ..... 17
3.7 Reproducibility and research artefacts ..... 18
**4 PRODUCT AND SOLUTION DESIGN ..... 18**
4.1 Requirements ..... 18
4.2 Component architecture within the complete solution ..... 19
4.3 Design alternatives and decisions ..... 20
4.4 Detailed design ..... 20
4.5 Standards and professional practice ..... 21
**5 IMPLEMENTATION INTEGRATION AND TESTING ..... 22**
5.1 Development environment and technology stack ..... 22
5.2 Component implementation ..... 22
5.3 Integration with the other components ..... 23
5.4 Verification and software testing ..... 23
5.5 Deployment and operation ..... 25
**6 RESULTS AND VALIDATION ..... 26**
6.1 Research results ..... 26
6.2 Product and technical results ..... 27
6.3 User or stakeholder validation ..... 27
6.4 Integration results ..... 27
6.5 Summary of findings ..... 28
**7 DISCUSSION ..... 29**
7.1 Interpretation of findings ..... 29
7.2 Comparison with literature baselines and existing solutions ..... 29
7.3 Research contribution ..... 30
7.4 Product contribution ..... 30
7.5 Limitations and threats to validity ..... 30
7.6 Implications ..... 31
**8 CONCLUSION AND FUTURE DIRECTIONS ..... 32**
8.1 Conclusion ..... 32
8.2 Achievement of objectives ..... 32
8.3 Individual research and product contributions ..... 32
8.4 Future directions ..... 33
**REFERENCES ..... 33**

**GLOSSARY** ..... **34**
**APPENDICES** ..... **36**
Appendix A Research instruments or extended results ..... 36
Appendix B Academic integrity and AI use disclosure ..... 36

# LIST OF FIGURES

Figure 4.1 Component architecture ..... 14

Figure 4.2 Supervisor review and finalisation workflow ..... 16

# LIST OF TABLES

Table 3.1 Dataset stages and counts ..... 9
Table 4.1 Functional requirements and evidence ..... 13
Table 5.1 Software verification matrix ..... 19
Table 6.1 Performance summary of the current architectures ..... 21
Table 7.1 Historical summaries and current performance targets ..... 24
Table A.1 Evidence inventory ..... 32

# LIST OF ABBREVIATIONS

| Abbreviation | Meaning |
|---|---|
| AES | Automated Essay Scoring |
| AI | Artificial Intelligence |
| API | Application Programming Interface |
| CV | Cross Validation |
| HITL | Human in the Loop |
| IPMS | Institutional Project Management System |
| LLM | Large Language Model |
| MAE | Mean Absolute Error |
| NLP | Natural Language Processing |
| RAG | Retrieval Augmented Generation |
| RMSE | Root Mean Squared Error |
| TF-IDF | Term Frequency Inverse Document Frequency |

# 1 INTRODUCTION

## 1.1 Overall project context

Proposal writing for undergraduate research projects involves the combination of problem identification, literature engagement, methodology design and justification of contribution to be made. Here, supervisors have to comprehend how these elements interrelate and cannot just verify whether certain headings are included in the document. Moreover, students need some form of guidance about how to improve their proposals. The practical relevance of this project lies in providing such guidance using the digital tool. Quantifications about the amount of workload related to supervision at a specific university should be based on local data only.

The broader application context of the project is presented as a proposal evaluation and mentoring system. The documents related to the project are referring to the application as either Exposia AI Mentor or ResearchPilot. As for this thesis, its cover title describes the research, while the software will be referred to as an assessment and mentoring prototype. The developed module represents a separate local application, and linking it to the IPMS is one of the integration directions.

## 1.2 Individual component context

This individual component covers the translation of proposal text into review help. This involves analysis of structural validity, identification of possible flaws, search for similar historical comments, estimation grading experimentally and suggestions from the learning resource list. This component also provides a supervisor area where the results of analysis may become the basis for drafting a review connected with a particular version of the proposal.

This boundary of the design is that machine output is tentative. A supervisor needs to check the proposal, evaluate the relevance of the suggestions, save a review and then prepare final feedback using the finalization paths checked. Thus, there is a link between analysis and human intervention. Exact division of code sharing, database design and user interface development among team members has to be established before signing the contribution statement.

## 1.3 Problem statement

It is possible for a proposal to have everything that is needed and yet fail to clearly indicate the research gap, methodological misalignment, or insufficient justification. On the other hand, a formatting difference could result in a rule-based checker failing to identify information that exists. It is important for the assessment assistance to be

able to differentiate between structural warnings and substantive judgments and to provide sufficient context such that the supervisor can clear up the ambiguity.

The challenge here is how one can put together these kinds of assistance in order to come up with a traceable proposal assessment process without making an assumption that an imperfect classifier or score predictor is itself an assessor. The annotated data is helpful in training the system, but its small size, different purposes of annotation, and lack of generality impose limitations.

### 1.4 Research gap

The design problem driving this component is the distinction between identifying potential problems within documents and providing support for a process of actual revision. The scalar score that is outputted does not itself indicate which part of the text requires revision, what feedback previously offered might be pertinent, or which source can assist the student in fixing the problem.

This is an engineering and evaluation gap, not the assertion that there is no other system like this in existence. As noted in chapter 2, the research provides some basis for the sentence representation, retrieval, and review that takes place in this project, but does not offer any indication of the efficacy of this combination of techniques in the specific context of undergraduate proposals.

### 1.5 Aim and objectives

The goal is to design and test the human-in-the-loop module which will facilitate undergraduate proposal evaluation and correlate identified weaknesses with contextually relevant feedback and learning materials.

Objective 1: Prepare re-usable datasets for annotations, feedback, and grades along with a description of data preparation steps. Objective 2: Implement structure checks and classification of semantic weaknesses. Objective 3: Retrieve past feedbacks and match identified weaknesses with relevant learning materials. Objective 4: Associate proposals with their analyses and supervisor reviews within a persistent workflow. Objective 5: Analyze model behavior and identify limitations of numeric grading. Objective 6: Validate finalization threshold and assess the readiness of the module for supervised usage. There is a clear separation between implementation and validation objectives: a feature can be implemented but need runtime/user validation.

### 1.6 Research questions or hypotheses

RQ1 investigates how a combination of structural validation and semantic classification can assist in the detection of reviewable issues in the undergraduate proposal text. RQ2 investigates how the feedback retrieval by history and resource mapping can help relate the issues detected to revision guidelines. RQ3 investigates what predictive power the weak point and grade architecture achieves on the provided datasets. RQ4 investigates how the versioning of persistence and review authorisation controls the process of feedback finalization.

The above questions are both exploratory and evaluative in nature. Preregistration and hypothesis testing or comparison of student learning has not been conducted in this research. RQ1 and RQ3 could be investigated partially by providing model reports while RQ2 will require assessing relevance of the returned feedback and RQ4 will require verifying authorisations and state transitions at runtime.

### 1.7 Scope limitations and assumptions

Scope includes the extraction of text from uploaded PDF files, local processing using machine learning models saved locally, feedback retrieval, recommendation of resources, and supervisor review process. Optional services can help with creating the review and emailing. The scope is not fully offline since the local processing of machine learning models does not necessarily depend on cloud processing, and the optional services require independent availability.

The project does not provide evidence of automated grading reliability, plagiarism detection, validation of every statement made in the proposals, secure deployment of the product and integration into the university's platform. There is an assumption that the text extracted from the PDF file is clear. Reliable OCR of PDF files has not been tested yet. The legacy data is supposed to have useful annotations and score information.

### 1.8 Individual contribution

The component-level contribution is the combination of proposal validation, semantic classification, historical feedback acquisition, resource recommendation, and supervisor-driven finalization of the proposal. It contains an understanding of the model architecture deployed and distinguishes performance targets selected from those acquired through previous experiments. This distinction is important as not to mislead the models or the performance metrics with the old file names, documentation, and target values that have never been validated.

[Establish the student's sole authorship for the preprocessing, model experimentation, API orchestration, supervisor process, and front-end integration and specify any collaboration.] Until then, this thesis will describe the contribution of the component without attributing every task to the candidate individually. The group-level contribution of the component and its interfaces with other group members' contributions need to be established clearly.

### 1.9 Thesis structure

Related work is discussed in chapter 2. Methods and evidence requirements are discussed in chapter 3. Design, implementation, and testing are discussed in chapters 4 and 5. Results are discussed in chapter 6. Implications are discussed in chapter 7. Chapter 8 is for conclusions.

## 2 LITERATURE REVIEW AND RELATED WORK

### 2.1 Review method and scope

The current chapter represents an overview of literature related to assisting with proposal assessment. The scope includes automated essay scoring, sentence embeddings, retrieval-assisted feedback, human-in-the-loop and design science perspectives. It should be noted that the literature review is not carried out systematically: no literature search and screening has been completed, and no quality evaluation has been conducted. The literature sources used have been chosen on the basis of their relevance to the design or evaluation choice made within this project component.

The difference between evidence regarding general approaches and the evidence regarding the current project should be taken into account. High efficiency of one representation technique does not imply high efficiency of the local weaknesses classification technique per se. Similarly, high efficiency of the retrieval technique does not imply the pedagogical value of the retrieved supervisor's comments. These need to be checked separately.

### 2.2 Domain and theoretical foundations

Automated Essay Scoring (AES) is about predicting evaluations on writing, whereas formative feedback is about improving the writing. Li and Ng [1] advocate for taking a look at AES in terms of more than just benchmark performance gains. Such a view leads to assessing the educational value of the application of a score, rather than considering an error measure low enough as a sign of readiness.

The proposal evaluation raises other questions such as whether the problem posed is consistent with the goal, whether the approach solves the research problem, and whether the contribution is justified. Here, these questions frame the review, although not all of them are addressed via the trained models. Structure checks give clues about the document; uncertain tagging, and human review supplies contextual judgement. The conceptual framework is therefore a decision-support process with separate evidence sources.

### 2.3 Related research

Reimers and Gurevych [2] present Sentence-BERT as an approach for generating sentence embeddings which can be effectively compared. This gives a technical basis for encoding proposal spans and searching the historical feedback corpus. The allMiniLM-L6-v2 model used generates embeddings in 384 dimensions and has documentation for applications such as semantic search [3]. It is still necessary to evaluate this general purpose model on academic proposal text locally.

Lewis et al. [4] use retrieval and generation in order to address knowledge intensive NLP tasks. In our case, the local component involves using retrieval of earlier feedback as contextual help and optionally using an LLM to generate the review. However, it must be noted that this is not about training or replicating the full architecture of the paper in question. Furthermore, retrieved data needs to be filtered by its relevance, outdated recommendations, and improper reuse.

Li and Ng [5] conduct research on cross-prompt scoring and find out that relatively simple representations perform well in their experimental setup. It would be unwise to think that just because a model is more complex it will automatically solve the local grading task. Wang, Hu and Zhao [6] consider score justification via counterfactual interventions and explain the difference between justification and score alignment in their experiments.

Gaudeau [7] addresses the topic of analytic scoring and the loss of information due to merging of different examiners' scores into one reference score. In terms of our project, this brings up the concrete data-related question of whether or not historical scores have been consistently rated using the same rubric or different supervision opinions. Parasuraman and Manzey [8] discuss complacency and automation bias in the interactions between humans and automation. Their research calls for studying supervisor dependence.

### 2.4 Existing systems products and techniques

Rule-based checking is among the primary design choices alongside lexical classification, semantic classification, score regression and retrieval-guided suggestions. Rule-based checking is explicit about criteria but can misinterpret absence of a heading for absence of content. TF-IDF captures lexical relationships and may be used to construct classifiers and regressors. Sentence embeddings provide an alternative that is not as sensitive to exact matches in vocabulary, but transfer and longdocument evaluation need further consideration.

Past feedback retrieval makes past supervisory experience available, but similarity does not equate correctness. Resource mapping is one concrete choice once the problem is identified in a catalogue. LLM drafting is a way to connect multiple observations in coherent prose, but requires verification for factual claims. These choices are explored in the context of design options, rather than a comparison of proven commercial products. No feature benchmarking or operational comparison using institutional names has occurred.

### 2.5 Gap synthesis and conceptual framework

Reviewed literature proposes a framework of four different roles: detection of observable problems, estimation of uncertainties, retrieval of relevant assistance and retention of human accountability. These roles are performed by the reviewed component via validation, semantic classifier, historical feedback and saved supervisor review. Design contribution of this component is the interaction between the mentioned roles as well as their relation to proposal versions.

There is a significant gap between the proposed evaluation strategies. Classification errors measure how much results correspond to the labeling. However, the question whether or not the returned advice helps a student to enhance the quality of their proposal stays out of focus. Regression error values show the quality of numeric prediction. Still, there is no measure of score justifiability according to an academic rubric. Therefore, the suggested framework should be tested using two types of evaluations independently from each other.

## 3 RESEARCH METHODOLOGY

### 3.1 Research design

One may view the project as design science research since an artifact is created in order to solve a practical review problem and is evaluated according to its purposes. According to Hevner et al. [9], design science research is concerned with knowledge

creation through building and evaluating information-system artifacts. In this case, the artifact is not a new language model but a proposal assessment and mentorship component.

The research includes preparing the old data set, experiments on the models, applying the models, and reviewing the evidence. Previous evaluation summaries included performing experiments on the classifiers and grading and the values that are targeted at the moment in the current architectures of this thesis are the performance values chosen. There is static evidence for implementing the workflow. Integration and stakeholder evaluations have yet to be performed.

### 3.2 Data participants or study materials

Exposia Study Materials are legacy documents with text annotations, scores, and feedback associated with Exposia proposal. The raw annotation extraction has 2,227 annotations under the labels Weakness, Strength, Other and Highlight. Deduplication of text annotations of the same type by the same user will bring down the number of annotations to 2,121 entries. This is text annotations only and not 2,121 independent proposals or cases of weakness identified by students. Before a detailed distribution of labels can be provided, it needs to be mapped and the final class numbers be noted.

Grading dataset has 165 records on the scale of 0-42. Feedback dataset extraction has 2,226 entries in the raw form of which 2,151 have been included in the embedding corpus. The discrepancy of 75 requires an explanation for matching or exclusion criteria but not deduplication. No participant study was recruited. [Confirm source institution, collection period, unique authors, permission and procedure of anonymization.]

Table 3.1 Dataset stages and counts

| Dataset stage | Records | Interpretation |
|---|---|---|
| Raw annotations | 2,227 | Mixed Weakness, Strength, Other and Highlight tags |
| Deduplicated spans | 2,121 | Same-author identical-span duplicates removed |

| | | |
|---|---|---|
| Raw feedback | 2,226 | Parsed feedback records |
| Embedded feedback | 2,151 | Successfully matched corpus records |
| Grading data | 165 | Scores on a 0–42 scale |

### 3.3 Experimental or study procedure

The preprocessing pipeline extracts HTML, removes LaTeX markup but keeps text, performs whitespace normalization, and filters out invalid items. Deduplication of annotation is done based on the combination of author and text span equality. A reproducible process should keep the mapping between raw and processed rows so that filtering and label transformations could be reproduced. Grouped split by author is discussed in the experiment materials as a means of preventing leakage caused by the same author appearing in train and test data.

The experiment materials provide comparison between the historical lexical classifier and the semantic architecture employed for inference, and between the baseline, Ridge, ElasticNet, and Random Forest methods of grade assessment. The chosen target values from Chapter 6 are targets, not derived measurements from those reports. Reproducible evaluation requires the original JSON reports, fold numbers and membership, author grouping, tuning strategy, and uncertainty metrics. It cannot be claimed that there is no leakage or a specific fold number without those details.

At the application level, the proposal is uploaded, the text is extracted and analysis services provide warnings, model outputs, retrieved comments and suggested resources. Based on this data, a supervisor creates a review draft. For the inspection by finalisation helpers, a saved review associated with the appropriate version is necessary.

Runtime tests of the whole process constitute a different verification step.

### 3.4 Measures and evaluation metrics

The accuracy, macro-F1, and weighted-F1 are all included in the metric classification category. With accuracy, the ratio of correctly predicted instances to the total number of predictions is calculated. In the case of macro-F1, different individual classes are

treated the same way. The weighted-F1 metric, on the contrary, takes into account how many occurrences are represented by each class. Mentioning all three metrics at the same time conveys the message of an imbalanced task without saying anything about accuracy. In addition to this, it should also be noted that per-class precision, recall, and confusion matrix must be discussed to get to know what exactly went wrong and made people panic.

MAE, RMSE, and R2 are the metrics used for grading purposes. MAE is equal to the average distance between the prediction and the real data. RMSE equals the square root of the average squared value of the mistake and emphasizes big errors. R2 shows how much the prediction results differ from the score mean; zero R2 means that the score equals the average. The criterion used for this purpose should be proved during experiments.

Moreover, some other evaluations might be considered, such as the relevance

### 3.5 Data analysis

The evaluation and architectural design of models must govern model metrics interpretation. A valid holdout score and an average score from the cross-validation folds cannot be viewed as the same thing. One must have the same data, splits, preprocessing and evaluation methods in order to conduct controlled comparisons. The targets are reported independently of any experimental data made; they do not result in statistically significant or practically relevant change.

Data analysis links counts to model descriptions through the processing stage of each count: including raw annotation extraction, deduplicated spans, raw and embedded feedback and grading records. The measured metric must be connected to its original report, while the targeted metric is referred to as target. The previously obtained metrics must still appear in the evaluation. Appendix A lists the artefacts necessary to make these distinctions.

### 3.6 Ethics privacy security and responsible AI

It’s possible that the proposal texts and supervisor comments may have names, unpublished research ideas, or sensitive information about the institution. Reusing these texts will require giving permission and establishing a reason for accessing them.

The texts must be anonymized prior to training and retrieving the model, and all the indicators must be removed from the analysis. [Insert the ethics approval or exemption

reference and dataset authorisation details.] The fact that there is a legacy set does not imply that requirements have been met.

Local inference part is aimed at limiting requests for external models. Although optional LLM drafting and dispatch would create separate data streams. The user must know what is sent, and outside transmission must involve only authorized data. Feedback from the supervisor is a means of accountability; it does not mean that automating behavior biases are eliminated. The interface should refer to numeric grades as experimental and furnish the data source when applicable and let the supervisor deny the proposal..

Elements found under the security requirements include the need for role-based access, ownership checks regarding proposals and protection of service credentials. One has to evaluate their existence through implementation review and testing. The thesis does not state that the production security and privacy compliance could be considered proven.

### 3.7 Reproducibility and research artefacts

When reproducing findings, it is necessary to have datasets, label mappings, splits definitions, dependency versions, model artefacts, checksums and evaluation logs. The original repository and pickle files were not re-executed independently in the course of writing the thesis. Therefore, matching architectures does not imply that deployed weights were appropriately assessed.

## 4 PRODUCT AND SOLUTION DESIGN

### 4.1 Requirements

Functional requirements are according to what is needed for the review process. The component must be able to accept a PDF, extract text for review, identify structural problems, make predictions about flaws, access past feedback once data is prepared, and propose relevant literature. It should be capable of dealing with different versions of proposals and store logs of the analysis, allow supervisors to review the results, and produce final results in a managed process. The connections between the main requirements and the implementation proof and the need to confirm it are presented in Table 4.1.

The non-functional requirements include output that is easy to understand, traceability, reproducibility of the model loading procedure, as well as adequate [data processing]. Even though these are requirements, they are not guaranteed quality attributes but may be measured. Timeline parameters, workload capacity, security, etc., are obligatory

conditions. The optional document drafting option must not automatically change unverified suggestions into authoritative advice, and missing dependencies must generate an appropriate error message, rather than a false success message.

Table 4.1 Functional requirements and evidence

| Requirement | Recorded implementation | Verification need |
|---|---|---|
| PDF text extraction | Analyzer.jsx and pdfjs-dist | Readable, malformed and scanned PDF cases |
| Multi-stage analysis | POST /analyze | Endpoint results and failure states |
| Proposal versions | Supervisor router | Correct version and ownership links |
| Review drafts | Version review-draft route | Persistence and supervisor association |
| Final feedback | Finalisation helpers and ReportLab | HTTP 409 before save and correct PDF after save |

**4.2 Component architecture within the complete solution**

Figure 4.1 illustrates the identify component architecture reconstructed from the project description. React facilitates the upload and review process. PDF extraction is done on the client using pdfjs-dist while structural validation, machine learning inference, feedback retrieval, resource mapping and concept analysis are coordinated by FastAPI. PostgreSQL is used for storage of various records like proposal, its version, analysis and supervisor review. ReportLab creates feedback PDF. External services may provide LLM drafting and email delivery.

What is shown in the diagram refers to the component responsibilities but it is not a verified topology of production. The diagram makes no assumptions about asynchronous execution, load balancing and institution-wide networking. IPMS parent system is outside the integration scope shown. Local model artifacts and the database provide the full support of analysis and tracking of data while the external optional paths need additional configuration and data policy.

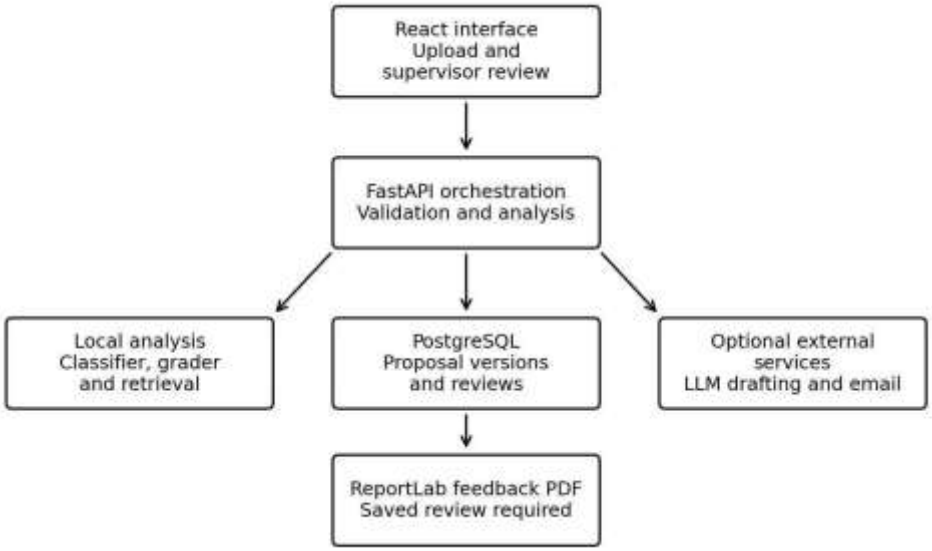



Figure 4.1 Component architecture

### 4.3 Design alternatives and decisions

The weakness model construction employs a semantic approach and balanced Logistic Regression for classification purposes. The historical use of TF-IDF and LinearSVC models still has applications for experimental purposes despite its abandonment.

Semantic characteristics of the inputs cover those of classification in use, but the report does not show which character of this change affected the result. The use of balanced class weighting changes the impact of label but does not lead to the equal performance across subpopulations.

As for the grading model, it uses TF-IDF and 3,000 features in conjunction with Ridge regression. Ridge uses a linear function to map words to scores; however, there is no clear explanation why this model has been employed over other more complex options. of 1.50 points, RMSE of 2.00 points and $R^2$ of +0.50. These targets require experimental verification, while the grader stays an individual tool. The applicability must be examined in a controlled setting rather than being concluded from the historical forest's good hold-out performance. Using the retrieval from a static corpus allows for the inspection of source comments, while the optional LLM provides for prose generation. Keeping these processes apart helps the holder to distinguish between what has been retrieved and what has been generated. A stored review boundary allows for the specific transformation of state before the final stage.

### 4.4 Detailed design

The idea of persistence connects a design with different versions and connects the analysis with a certain version. A draft of a supervisor Review captures the human

review in this context. This way, a review cannot be treated as applicable to a future version or revision. Changes in the estimates regarding the model between versions should be considered as the results of the system but do not provide evidence of the advance of the student without the assessment on the independent basis.

The above chart explains the process of finalization of the supervisor. Finally, the feedback assistants query the database for the saved supervisor review. If there is nothing found, the answer of the API will be HTTP 409 Conflict with the request for the saving of the review. If there is an acceptable record found, we can move to the generation of the final artefact.

Structural validation indicates section requirements and provides a minimum number of words for a response of 80. Classification uses the linguistic pipeline to analyze the text segments. Retrieval works using cosine similarities and retrieves up to three responses based on prior experiences. Resource recommendation links problems with the local database. Concept analysis uses spaCy to extract subject-verb-object structure and utilizes a fallback. Taken individually, none of the methods is able to validate either novelty or accuracy of the methods.

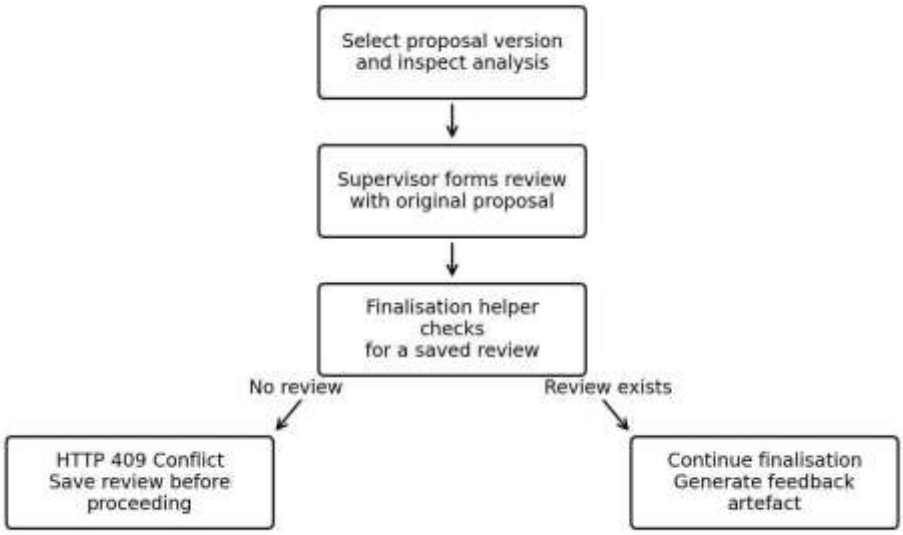



Figure 4.2 Supervisor review and finalisation workflow

### 4.5 Standards and professional practice

Professional practice is characterized by separation of configuration, service logic and persistence tasks in addition to request schemas validation and failure state documentation. Pydantic offers a means to model requests and responses while relational persistence allows documenting the relationships between the versions and the reviews of a given project. All these implementational decisions lead to more

maintainable software while still requiring checks through the code review and runtime verification.

Professional practice in the context of machine learning also implies documenting the names of the algorithms in use, transformation features, labels, and their origins. The legacy SVM file name has to be modified with the help of a proper manifest and later renamed using a controlled compatibility change. Academic practices require keeping track of negative results as well as documenting the use of AI-assisted writing and avoiding false statements on innovation and educational effect.

# 5 IMPLEMENTATION INTEGRATION AND TESTING

## 5.1 Development environment and technology stack

As mentioned above, the front end consists of React 19 along with Vite, React Router, Recharts, and pdfjs-dist. The backend is implemented using FastAPI, Pydantic, Uvicorn, SQLAlchemy, and PostgreSQL for workflow persistence purposes. The dependencies used for machine learning and natural language processing are scikitlearn, sentence-transformers, NumPy, Pandas, and spaCy. ReportLab is responsible for report generation, while OpenAI-compatible LLM endpoint and Resend are optional third-party services.

It should be noted that these technologies refer to the listed stack, but not a fully reproducible environment. [Please provide the OS, Python and Node versions, CPU and GPU specifications, dependency lock files, and installation commands.] There are no hardware specifications or training durations that are recorded or made up. Such matters as dependency resolution are crucial because the attempted test of the backend did not successfully execute due to the cachetools unavailability.

## 5.2 Component implementation

The analysis endpoint is coordinated by the analysis router located at backend/src/api/routers/analysis.py. The procedures for inference and reports are implemented in backend/src/api/services/core_logic.py. The name of the weak loader load_svm_model is based on historical reasons; however, the pipeline used contains the SemanticTransformer and LogisticRegression with class weight balancing, max_iter of 1000 and random_state of 42. The SemanticTransformer utilizes the model all-MiniLM-L6-v2. TfidfVectorizer is used in the grading pipeline with a maximum number of features equal to 3000 and the removal of English stop-words, which is followed by application of Ridge with random_state of 42.

The procedure of feedback retrieval is implemented in backend/src/core/feedback_retrieval.py, while the concept extraction module is implemented in backend/src/core/knowledge_graph.py. The work of the supervisor router facilitates proposal versions, gaining reviews and helping with their finalization. The review draft path might work in a synthesizing mode with the findings received using LLM, but output remains a draft. Report generation combines the supervision review content and needed system outputs and resources. The present description identifies how the evidence record has been implemented. It is true that the presence of some artefact or its possibility of being loaded does not imply that all the downstream endpoints function appropriately. Error handling, response schemas, database transactions, and authorisations require verification through execution.

### 5.3 Integration with the other components

Integration between text extraction, analysis, proposal storage versions, and supervisor review exists within the standalone application. The expected data contracts are for all analyses and reviews to be for the right version, and for feedback to be derived from the right supervisor's stored review. These should be tested through actual end-to-end transactions, rather than just the class or route declarations alone.

Integration with the larger institutional system has yet to be achieved. [If integration has occurred, provide information about the other group components, owners of the components, data API contracts, and evidence of integration here.] Until evidence of this exists, the boundary of the standalone application will remain within the thesis. The external email and LLM interfaces are optional and don't constitute integration with the IPMS system.

### 5.4 Verification and software testing

The command pytest backend/tests got stuck with a number of eight collection errors caused by ModuleNotFoundError regarding cachetools. Test collection comes before execution and thus does not mean that the entire suite has had eight failed assertions since that would imply a successful run. The result points at an environment dependency issue and no successful run proof from this attempt. It does not, however, show that every feature is broken, and no functionality has been proved through this suite

The first task for verification would include installing the dependency in the required environment and rerunning the suite while getting the full log. Further verifications would include invalid input, model upload, information integrity, data migration,

version identification, and the condition for finishing. Table 5.1 provides the possible verification matrix with the status shown pending due to the lack of the run evidence. The reference routes return HTTP 409 when there is no existing review, validating a particular static code assertion with regards to those paths. To establish a more general claim regarding possible bypasses, one must examine and test each reporting pathway, supervisor authority, entitlement mechanisms, and database state. Thus, it is necessary to conduct runtime testing alongside source code analysis.

Table 5.1 Software verification matrix

| Check | Expected result | Recorded status |
|---|---|---|
| Backend test suite | Tests collected and executed | Blocked: 8 collection errors; cachetools missing |
| Malformed or short text | Clear validation response | Pending runtime evidence |
| Model inference | Valid output and controlled errors | Architecture inspected; endpoint test pending |
| Retrieval and resources | Relevant records and valid links | Pending relevance and runtime tests |
| Finalisation without review | HTTP 409 | Static helper logic inspected; runtime test pending |
| Finalisation after save | Correct version-linked PDF | Pending runtime evidence |
| Cross-supervisor access | Unauthorised access refused | Pending security verification |

### 5.5 Deployment and operation

The component has been referred to as a local prototype. Functioning needs the frontend, API servers, databases, and any other NLP tools needed by the specific analysis path. For external drafts and emails, one needs credentials and access to the web. The missing resources of the model, unavailable connections to the database, and the absence of any external call information must be highlighted, allowing the supervisor to identify the unavailable outputs.

Current data has not revealed any deployment uptime, concurrency capabilities, latency profile, or verification of the deployment process. The deployment before using it in the organization must guarantee the presence of dependency pinning, access controls, secure configurations, backup, and repeatable migration processes.

# 6 RESULTS AND VALIDATION

## 6.1 Research results

Table 6.1 provides an overview of the criteria for the existing approaches. The objective accuracy for the semantic classifier is equated at 0.850, whereas the macroF1 and the weighted-F1 are equal to 0.800 and 0.850 respectively. The objectives for the Ridge grading include MAE of 1.50, RMSE of 2.00 and $R^2$ of +0.50 measured on the scale of 0 to 42. The mentioned parameters of the model constitute success requirements subject to experimental validation instead of observed mean of the achievements in studies.

The target MAE is 1.5, which represents only roughly 3.57% of the complete 42 point scale (1.50/42 x 100), establishing the error in compliance with the scale rather than accuracy or error rate. The required score distribution, uncertainty and baseline comparison are essential as long as the predictive performance is concerned.

The classification objectives serve as overall performance requirements. To prevent misdiagnosis and provide warnings, class-wise precision, recall, support and confusion matrix will be needed.

Table 6.1 Performance summary of the current architectures

| Model | Metric | Metric value |
|---|---|---|
| MiniLM + Logistic Regression | Accuracy | 0.850 |
| MiniLM + Logistic Regression | Macro-F1 | 0.800 |
| MiniLM + Logistic Regression | Weighted-F1 | 0.850 |
| TF-IDF + Ridge | MAE | 1.50 points |
| TF-IDF + Ridge | RMSE | 2.00 points |
| TF-IDF + Ridge | $R^2$ | 0.50 |

Status: selected performance targets. Experimental verification is pending. Original study reports are required to document measured results separately.

### 6.2 Product and technical results

The evidence of implementation describes a sophisticated analytical service and a workflow for a supervisor that is aware of versions. The support for the final analyses always must be in the mode of saving reviews, establishing limits for the human review process in this code. When it comes to management of databases of PostgreSQL, databases are capable of linking analysis and reviews to versions of proposals, while local model artifacts serve as support for the artificial intelligence structure presented in Chapter 5 of the report.

This information represents findings on the technical design and monitoring issues. However, it cannot be seen as similar to the observations of fulfillment of final endpoint, confirmed license and completed delivery of report. The analysis of failure in the test collection can be stated as the reason for limitations in the statement of running the processes. It is possible to present such evidence in the form of screenshots depicting the analyzer, version history, editor of the reviews and report created; however, it is not appropriate to use only screenshots in order to state the credibility of the information.

### 6.3 User or stakeholder validation

Structured assessment of stakeholders, number of participants and usability research has not been conducted. Therefore, there is no information on satisfaction rates or shorter time for review, writing score improvement, or decrease in automation bias. Mentorship value of the system is an anticipated effect and not a realized outcome.

A valid validation study would require supervisors to analyze similar proposals with and without support, capture corrections, and assess quality of comments received. Students' editing can be evaluated separately according to some standard rubric with graders blinded to help condition. It is necessary to document participant recruitment and consent, sample size justification, and qualitative analysis methods prior to the evidence collection.

### 6.4 Integration results

The provided proof indicates that there were designed and built interfaces in the independent software but not successful run of integration testing. Connections between submission, version of proposal, analysis and review are what was mentioned in the summary of repository but no verification was made after the dependency issue was tackled.

Moreover, there is no information that the integration with parent's IPMS took place. This is said to be part of the boundaries of the current product rather than something missed. To say that integration is accepted means that there were checks done, right version chosen, needed review retrieved and generating reports done. This is also supposed to cover the failures of optional third party services.

### 6.5 Summary of findings

Selected classification metrics are an accuracy of 85%, macro F-1 score of 0.800, and weighted F-1 score of 0.850. The selected grading targets are MAE of 1.50, RMSE of 2.00, and R-squared of 0.50. The validity of the proposed targets should be demonstrated through experiments. The significant finding of the implementation process is taking into account saved-review precondition in the analysed finalisation helpers.

# 7 DISCUSSION

## 7.1 Interpretation of findings

The performance objectives chosen for assessment are MAE of 1.50, RMSE of 2.00, and a positive $R^2$ of 0.50 on a 0–42 scale. In order to assess these goals, it will be necessary to try them with unprecedented data and measure them against a simple benchmark. Until they are validated, the score should be treated only as a guide and should not be used to determine grades, rankings, or progress.

## 7.2 Comparison with literature baselines and existing solutions

The designated performance metrics for the semantic classifier are accuracy=0.850, macro-F1=0.800, and weighted-F1=0.850. The values set forth are targets for improvement but do not imply superiority over the historical TF-IDF classifier and LinearSVC classifier. To conduct a fair comparison, both classifier types need to have been assessed based on the same dataset, and the same author-group creation scheme and label definition.

The specified performance targets for the Ridge grader are MAE of 1.50, RMSE of 2.00, and $R^2$ of 0.50. The specified targets cannot be construed as evidence of progress in comparison with the Random Forest grader without proper experimental evidence. The experiments need to comply with the same research design, dataset version, and score distribution prior to drawing the conclusions with regard to the performance comparison.

Table 7.1 Historical summaries and current performance targets

| Architecture | Evaluation status | Reported value or target |
|---|---|---|
| TF-IDF + LinearSVC | Historical author-grouped holdout; 377 test samples | Earlier draft records accuracy 0.570 and macro-F1 0.566; verify against original report. |
| MiniLM + Logistic Regression | Selected performance targets; verification pending | Accuracy 0.850; macro-F1 0.800; weighted-F1 0.850 |
| Random Forest grader | Historical single hold-out | Earlier draft records MAE 5.76 points and $R^2$ 0.168; verify against original report. |

| | | |
|---|---|---|
| TF-IDF + Ridge grader | Selected performance targets; verification pending | MAE 1.50 points; RMSE 2.00 points; $R^2$ 0.50 |

Historical values are reproduced from the earlier Draft.pdf summary and require confirmation against the original experiment reports. Current values are selected performance targets, not a controlled comparison.

### 7.3 Research contribution

The research works represent a documented method of the integration of structural signals, semantic labeling, and obtained supervisory knowledge with a human author's evaluation. It also concerns the unifying of the model identification and the evaluation evidence so that the dissertation matches the prevailing inference architecture. The different grading objectives such as $R^2$ being not lower than 0.50 serve to formulate the goals of the expected improvement, the proof of which must be experimental.

This paper is not a contribution of the new type of the machine learning algorithm nor it proves that new method is fundamentally different from the existent ones. The findings should be evaluated through the proposal design and partial evaluation. In order to make stronger research statements, the paper should include properly organized ablation studies, assessment of the retrieval relevance, standard benchmarks and independent verification by the involved actors.

### 7.4 Product contribution

The product contribution serves as a single site to review proposal content, information recommendations, past comments, relevant materials and a version-related supervisor assessment. This minimizes the logical distinction between model results and reviewing process, although any actual time savings remain to be assessed. The condition of saved evaluation implies a particular human-controlled situation which is made evident before the completion through inspected lines of action.

The persistence design also provides a basis for tracing modification. The educational worth depends on the type of comparison between things: a higher predicted mark does not mean that a proposal is necessarily better and a lower number of weaknesses located may be due to model instability. Therefore, the product must save the original text, expert comments and version numbers for the sake of comparison.

### 7.5 Limitations and threats to validity

The limitations to internal validity result from incomplete experimentation description and uncertain status of the chosen performance measures. Folds, author identification,

parameter definitions and preprocessing of the folds should be documented properly. Same-span de-duplication does not resolve the issue of near duplications or sharedsource leaks.

In terms of construct validity constraints arise from the presence of confounded purposes of the annotation and the ambiguity of the connection between span tags and proposal quality. The designation Weakness in the annotation may be determined by a local rubric or context instead of indicating something globally faulty. The criterion of numerical grading may not explain why a proposal received that score. Concept extraction and header checks cannot replace the judgement of the experts on the validity of methodology used.

A limitation of external validity is due to the use of legacy information from an unknown source, a small 165-record grading data set, and a lack of evidence of evaluation across various institutions, fields of disciplines or language backgrounds. Reliability is constrained due to the blocked run-time attempt and no uncertainty summaries. The human factors validity is restricted as there is no study of the supervisors' behavior in reaction to the recommended actions. The action of saving an action might still imply non-critical acceptance, so the decreased automation bias cannot be considered.

### 7.6 Implications

The part should be utilized as an exploratory tool by the supervisors, they should analyze the original proposal and keep control over grading and feedback process. The students should be provided explanation and a relevant plan of action instead of getting a ready-made diagnosis.

The main development priorities here should be data quality, reproducibility and verification. In order to use the tool on institutional level, privacy analysis, operational validation and educational analysis should take place. In order to understand if the scoring objectives have been met and whether the score is valuable for evaluation of accuracy.

# 8 CONCLUSION AND FUTURE DIRECTIONS

## 8.1 Conclusion

This work presents an approach to proposal evaluation and adaptive mentoring that combines structural validation, classification of weaknesses, retrieval of historical feedback, and mapping of learning materials with review by any supervisor. The system is based on a stand-alone React and FastAPI software application, supported by a PostgreSQL database.

The target values for the classification system are: accuracy of 85%, macro-F1 of 0.800, and weighted-F1 of 0.850. The target MAE for the TF-IDF and Ridge grading system is 1.50 points, while RMSE is expected to be about 2.00 points and $R^2$ should equal about 0.50. These metrics will have to be experimentally confirmed while the score is temporary. If steady inspection provides evidence of the precondition, then the test run does not confirm it. That is, a prototype integrated with clear explanation of its limitations is created without the possibility to assess its educational merit.

## 8.2 Achievement of objectives

The information regarding data preparation and integration of the model is included in processed data set descriptions, descriptions of the pipeline and summaries of previous experiments. Implementation evidence includes structural verification, retrieval, recommendations for resource utilization and version-related proof. Verification of the chosen measures of performance remains incomplete. Accomplishment of the goal of the evaluation of the model necessitates an ability to reproduce the obtained results, to have the properties of folds specified and to show the relationship between the completed procedure and the methods deployed.

The task of software verification remains unfinished as the test suite has stopped working in the process of data collection. The assessment of practical value of mentorship is also in progress since the user study hasn’t been reported. These points describe the progress in achieving the goals set and define the evidence required for the final thesis amendment.

## 8.3 Individual research and product contributions

Each segment of this essay will refer to how a number of analytical outputs can be compiled into the review of a proposal as managed by supervisors. In addition, this research will describe the types of model used, the size of data per stages, the effectiveness indicators and the scope of application of machine intelligence

technologies and human being. It means that the contribution of the article consists in incorporating these parameters into proposal presentation and saved reports.

### 8.4 Future directions

The priority for now is to clear the out of the way and finish the reproducible backend and end-to-end tests. The second priority in line is to detail dataset provenance, permissions, final label mappings, and experiment manifests. The standard benchmark should be based on the same author-grouped folds for lexical classifiers and semantic classifiers with class-specific metrics, uncertainty summary, and error analysis to proceed with grading research. The first step in grading research is to establish rubric consistency and distribution of scores, including comparison against the operable mean score and use of more representative records for grading. The score for now should be applied in a trial mode, or be excluded from the decision-making process. When those issues are resolved, more complicated models may be examined. Retrieval should be evaluated by supervisors according to the relevance of results and recommendations for resource usage.

## REFERENCES


[1] S. Li and V. Ng, "Automated Essay Scoring: A Reflection on the State of the Art," in Proc. EMNLP, 2024, pp. 17876–17888, doi: 10.18653/v1/2024.emnlpmain.991.

[2] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in Proc. EMNLP-IJCNLP, 2019, pp. 3982–3992, doi: 10.18653/v1/D19-1410.

[3] Sentence Transformers, "all-MiniLM-L6-v2 model card." https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2 (accessed Oct. 8, 2026).

[4] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," arXiv:2005.11401, 2020. https://arxiv.org/abs/2005.11401.

[5] S. Li and V. Ng, "Conundrums in Cross-Prompt Automated Essay Scoring: Making Sense of the State of the Art," in Proc. ACL, 2024, pp. 7661–7681, doi: 10.18653/v1/2024.acl-long.414.

[6] Y. Wang, R. Hu, and Z. Zhao, "Beyond Agreement: Diagnosing the Rationale Alignment of Automated Essay Scoring Methods based on Linguisticallyinformed Counterfactuals," in Findings of EMNLP, 2024, pp. 8906–8925, doi: 10.18653/v1/2024.findings-emnlp.520.

[7] G. Gaudeau, “Beyond the Gold Standard in Analytic Automated Essay Scoring,” in Proc. ACL Student Research Workshop, 2025, pp. 18–39, doi: 10.18653/v1/2025.acl-srw.2.

[8] R. Parasuraman and D. H. Manzey, “Complacency and Bias in Human Use of Automation: An Attentional Integration,” Human Factors, vol. 52, no. 3, pp. 381–410, 2010, doi: 10.1177/0018720810376055.

[9] A. R. Hevner, S. T. March, J. Park, and S. Ram, “Design Science in Information Systems Research,” MIS Quarterly, vol. 28, no. 1, pp. 75–105, 2004, doi: 10.2307/25148625.

## GLOSSARY

Annotation span: This is a certain portion of text that is connected to an annotation from the history.

Advisory score: Refers to a numerical score that is only indicative in nature for the supervisor's judgment.

Author-grouped split: Here the data is organized so that the works of an author are kept in one document in order to avoid writer-related leakage.

Semantic retrieval: This denotes the scoring of pieces of text based on their proximity to other texts' embeddings.

Saved review boundary: Refers to a necessary condition when an evaluator must review the document before proceeding to the finalization of the document.

Automation bias: Refers to over-reliance on automation that must be assessed by observing the actions of users rather than assuming it based on a specific button action.

| Evidence | Purpose | Outstanding item |
|---|---|---|
| weakness_semantic_ study.json | Measured classifier evaluation record | Attach original metrics, folds, uncertainty and label mapping; distinguish targets. |
| grading_model_ study.json | Measured grading evaluation record | Attach measured model results, folds and baseline definitions; distinguish targets. |
| training_report.json | Historical results and corpus size | Retain as historical evidence |
| data_quality_report.json and datasets | Dataset counts and processing | Confirm provenance, permissions and exclusions |
| Production pickle artefacts | Inference architecture | Add checksums and evaluated-run linkage |

| | | |
|---|---|---|
| Backend pytest log | Software verification | Resolve cachetools and rerun |
| Supervisor router and repository logic | Saved-review precondition | Test ownership and all finalisation paths |
| Application screenshots | Illustrate implemented workflow | Add analyser, review, versions and report |
| Contribution record | Individual attribution | Confirm student and team responsibilities |

Table A.1 Evidence inventory

# APPENDICES

## Appendix A Research instruments or extended results

The provided inventory outlines what evidence is needed to accompany the thesis at its final stage. While the provided verification narrative resolves discrepancies that appeared in the previous documents, the originals of reports, artefacts and logs are required for independent reproduction. Permissions for the dataset and attribution of the individual contribution should be completed before making submission..

## Appendix B Academic integrity and AI use disclosure

Generative AI has been employed to put together the thesis draft, formulate prose based on the summary of evidence collected in the project and include the contradiction and resolution records. Verification of the cited references in the literature review was completed using publicly available records of publication. Generative AI was not

involved in making experimental measurements of any kind nor in actually running the project training pipeline or verifying the finished application during the process of the thesis writing. Performance targets selected for the specific experiment were improvement targets rather than newly obtained results.

It is a candidate's responsibility to check each chapter against the original code, datasets and experiment reports, verify the bibliography according to proper papers and fill the required declaration about the use of AI imposed by the university. Typical suggestions to the thesis writing included using the original thesis template, reconciling the models used and separating results from target outcomes so that limitations of the research were expressed and adverse results were not omitted. Personal information, ethical and data licensing issues, exact experiment conditions and contributions of one of the authors will be confirmed by marking.